University of Exeter

MTH1002

Mathematical Methods

On Campus Exam	Closed Book
Additional Materials: None	**Permitted Materials:** Approved Non-Programmable Scientific Calculators Permitted (Green)

Guidelines:

-The **duration** of this paper is 2 hours
-Word Count: No Word Count
-Write all answers in the answer booklet provided.
-Do not open the exam paper until instructed by the Invigilator.
-Use only black pen or pencil.
-Mobile and electronic devices are not allowed.
-Do not communicate with other candidates during the exam.

Exam Instructions:

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

SECTION A

1. (a) Consider the function $f(x, y) = x^2 + y^2 - 4xy$.

   (i) Find the critical points of $f(x, y)$.

   (ii) Classify each critical point as a local minimum, local maximum, or saddle point. (10)

   (b) Express the definite integral

   $$\int_1^3 (9x - 4)\, dx$$

   as a Riemann sum and evaluate it. Verify your answer by calculating the integral using the rules of integration. (10)

   (c) Let $f(x, y) = e^{xy} + \sin(x)$. Find the Taylor series expansion of $f(x, y)$ around $(x_0, y_0) = (0, 0)$ up to the second-order terms. (10)

   (d) Let R be the region that is bounded by the curve $y = x^2$ and the line $y = 4$.

   (i) Sketch the region R.

   (ii) Evaluate the double integral of $f(x, y) = 6x^2 + 2y$ over R. (10)

   (e) Solve the following differential equations:

   (i)
   $$\frac{dy}{dx} = \frac{x + 4}{\sqrt{xy}}, \quad y(1) = 1.$$

   (ii)
   $$\frac{dy}{dx} + 2y = x, \quad y(0) = 1.$$

   (10)

[**50**]

SECTION B

2. (a) Use row operations to put the following system of linear equations into row echelon form (k is a constant value):

$$\begin{aligned} x + y + z &= 3, \\ 2x - y + kz &= 1, \\ x - y + (k-1)z &= 2. \end{aligned}$$

Hence, solve the system of equations. Leave your answer in terms of k. Discuss the nature of solutions for different values of k. (12)

(b) Use Lagrange multipliers to find the maximum area of a rectangle bounded by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

(13)

[25]

3. (a) A function F is defined by the formula

$$F(x) = \int_0^{x+e^x} e^{t^2}\, dt.$$

Write down an expression for the derivative $\frac{dF}{dx}$. Do not attempt to solve the integral. (6)

(b) M is a 2×2 matrix which has the property that $M^2 = I$ and $M \neq \pm I$, where I is the 2×2 identity matrix. For any two-component vector $\mathbf{x} \neq \mathbf{0}$, show that the vectors $\mathbf{x} + M\mathbf{x}$ and $\mathbf{x} - M\mathbf{x}$ are eigenvectors of M. Also, write down the eigenvalues of M. (7)

(c) Consider the differential equation

$$my''(t) + cy'(t) + ky(t) = 0$$

where $m > 0, c \geq 0$, and $k > 0$.

(i) Determine the value of c that leads to a non-decaying solution. For this value of c, what is the frequency of oscillation of the system?

(ii) Prove that the system decays fastest when $c^2 = 4mk$. (12)

[25]

4. (a) Consider the matrix

$$A = \begin{pmatrix} 1 & 11 & 111 \\ 1111 & 11111 & 111111 \\ 1111111 & 11111111 & 111111111 \end{pmatrix}.$$

Let $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ be a column vector of unknowns and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ be a column vector of constants.

(i) By using row operations, evaluate the determinant of A.

(ii) Hence, write down the possible number of solutions to the system of equations

$$A\mathbf{x} = \mathbf{b}.$$

(10)

(b) Use the fact that

$$\frac{d}{dx}\big(\sec(x) + \tan(x)\big) = \sec^2(x) + \sec(x)\tan(x)$$

to evaluate the integral:

$$\int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{\sqrt{x^2 - 2x + 2}}\, dx.$$

(15)

[**25**]

[END OF PAPER]